Building Agentic Workflows with AmSC MAG and IRI

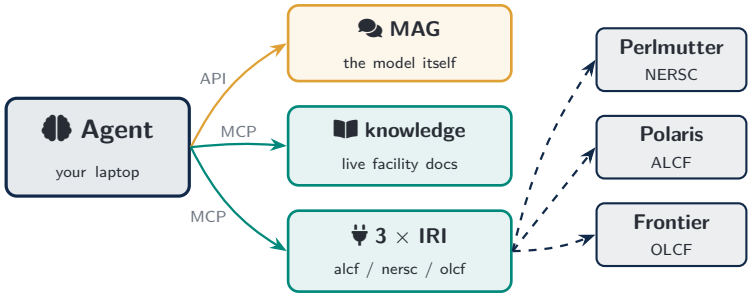
Genesis Mission October Hackathon

Huihuo Zheng
AmSC AI Services
Argonne Leadership Computing Facility
Argonne National Laboratory
October 14, 2026

huihuo.zheng@anl.gov

github.com/zhenghh04/gm_hackathon_agentic_ai

One agent reaches three facilities — through MAG, live docs, and IRI



Knowledge, reach, data — and judgment. The fourth leg is the `submit-job` skill.

Four MCP servers and one skill are the whole implementation

what you cloned

```
gm_hackathon_agentic_ai/
 .mcp.json              4 servers, one file
 mcp/
  auth_env.py           tokens, Globus id
  knowledge/
   docs_server.py        1 tool  AskALCF
  hpc/
   alcf_server.py       23 tools Polaris
   nersc_server.py      20 tools Perlmutter
   olcf_iri_server.py   17 tools Frontier
   *_iri_client.py      REST + Globus calls
 .claude/skills/
  submit-job/SKILL.md   the chain
 scripts/auth/          one login per site
```

The shape to copy

- **Server declares, client calls.** `*_server.py` is tool names and schemas; `*_iri_client.py` does the HTTP and Globus work.
- **Auth lives in one file.** `auth_env.py` reads `.env`, and resolves your laptop's Globus collection for `local_endpoint`.
- **Three facilities, one shape.** Same tool names on `alcf` / `nersc` / `olcf` — learn one, use all three.

Two folders, the whole system — `mcp/` for reach, `.claude/skills/` for judgment.

One sentence becomes a MNIST training job on a Polaris GPU

You

"Please run a MNIST training on Polaris using <ALLOCATION> on 1 node, debug queue."

write → stage → submit → monitor

```
# from one sentence, the agent writes code:
Write("train.py") # PyTorch MNIST, A100-ready
Write("run.sh") # #!/bin/bash -l, module load
globus_mkdir(dst=".../mnist/") # make folder
globus_transfer(src="train.py", dst=
  "/eagle/<project>/<you>/mnist/")
globus_transfer_status(task_id=...) # SUCCEEDED
submit_job(system="polaris", nodes=1,
  queue="debug", walltime="00:15:00",
  account="<ALLOCATION>")
get_job_status(job_id=...) # poll, backoff
read_file(path=".../job.out", tail=20)
```

What the job printed

```
Epoch 1/5 loss 0.42 acc 95.1%
Epoch 2/5 loss 0.15 acc 97.3%
Epoch 3/5 loss 0.10 acc 98.0%
Epoch 4/5 loss 0.07 acc 98.4%
Epoch 5/5 loss 0.05 acc 98.6%
Final test accuracy: 98.6%
```

Illustrative `job.out` tail — numbers vary run to run.

Where the rest of this lives

The full hands-on deck — 27 slides

Setup for all three agent CLIs, facility authentication, and three worked examples end to end.

drive.google.com/file/d/1exfjXwKcps5zbcO7kTGmXupfTkJJvh2q

The repository — clone and run it

The four MCP servers, the `submit-job` skill, and the step-by-step labs — everything here runs from a laptop.

github.com/zhenghh04/gm_hackathon_agentic_ai

Acknowledgement

This research used resources of the Argonne Leadership Computing Facility, a U.S. Department of Energy (DOE) Office of Science user facility at Argonne National Laboratory and is based on research supported by the U.S. DOE Office of Science-Advanced Scientific Computing Research Program, under Contract No. DE-AC02-06CH11357.